Basic (your first LCEL chain) — Solution

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Story — `lcel_chain_basics_practice.py`: this 3-piece chain — prompt, model, parser, joined with `|` — is the atom every LangChain thing you'll ever build is made of. Written here, once, on its own, so the `|` syntax and what each piece does is obvious before Doc06's tools, Doc08's RAG pipeline, or this document's own Build Task pile more onto it. **If not:** the first time you'd see `prompt | model | parser` would be inside a bigger chain already doing three other things, making it hard to tell which part the `|` syntax actually changed.

Solution

```
# lcel_chain_basics_practice.py
from langchain_core.prompts import ChatPromptTemplate
from langchain_openai import ChatOpenAI
from langchain_core.output_parsers import StrOutputParser

def build_chain(model_name: str = "gpt-4o-mini", temperature: float = 0.0):
    # why: model name and temperature as real parameters, not hidden
    # defaults — a model swap should be a visible, one-line decision.
    prompt = ChatPromptTemplate.from_template("Answer: {question}")
    model = ChatOpenAI(model=model_name, temperature=temperature)
    parser = StrOutputParser()
    # how: | connects the three pieces into one Runnable — the prompt's
    # output feeds the model, the model's output feeds the parser.
    return prompt | model | parser

if __name__ == "__main__":
    chain = build_chain()
    result = chain.invoke({"question": "What's 5 + 7?"})
    print(f"Answer: {result}")
```

Expected output:

```
Answer: 5 + 7 = 12
```